Tailoring Machine Learning Weather Predictions for Local Impacts

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Advancements in ML Weather Forecasting have made Tailored Forecasting Models Possible

- Traditional physics based weather forecasting models are expensive and slow to run. New ML based models can be run on a laptop.
- ML models are still very expensive to train and try to predict the entire globe at once.
- Can we improve accuracy on just the most relevant weather variables if we focus on a specific region?

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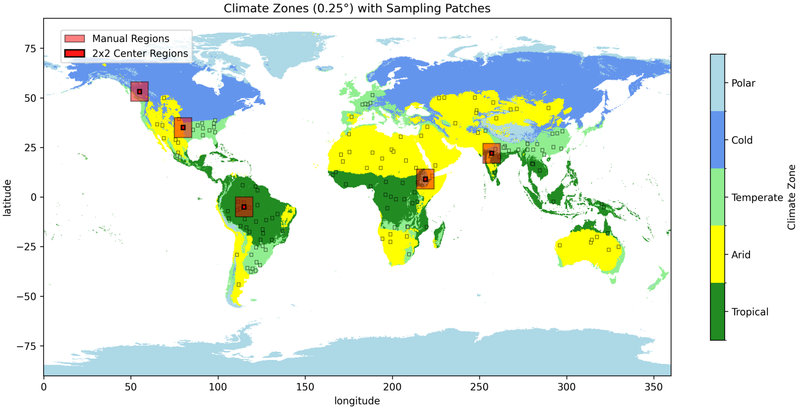
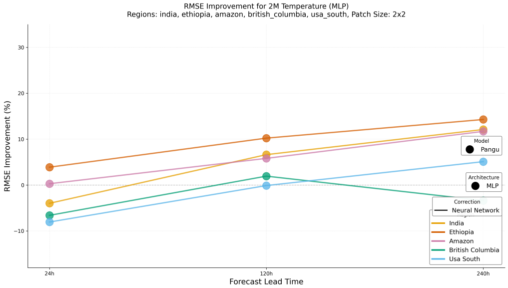
Apply neural network post-processing to both traditional and ML weather models, showing that it is possible to improve the accuracy of both (with perhaps some caveats).

Start by Manually Looking at Four Regions of Interest

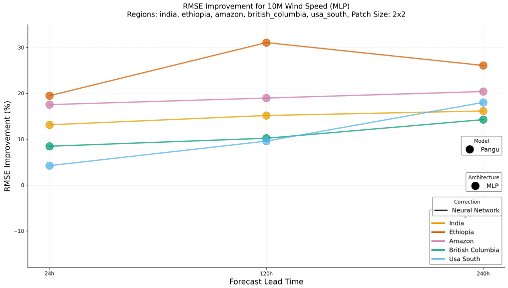
- Choose a handful of regions with a variety of climates and land types.



See Greater Improvements for Longer Lead Times and for Widn Speed

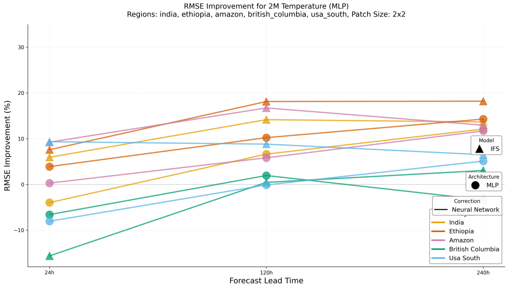


(a) 2m Temperature: Pangu (ML Model)

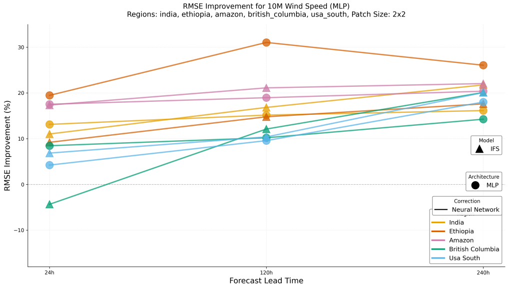


(b) Wind Speed: Pangu (ML Model)

NWP Model Can be Corrected In a Similar Manner

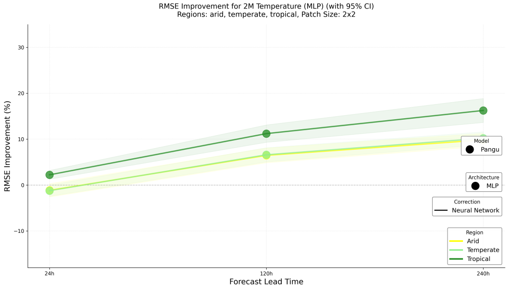


(a) 2m Temperature: Pangu (ML) + IFS (NWP)



(b) Wind Speed: Pangu (ML) + IFS (NWP)

Climate Zones



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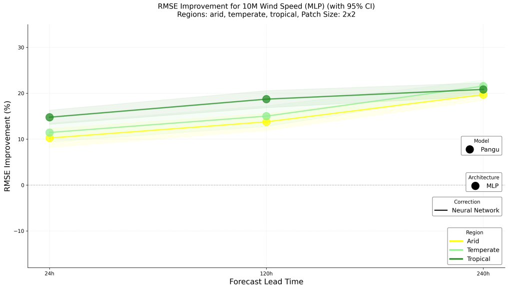

(b) Wind Speed: Pangu